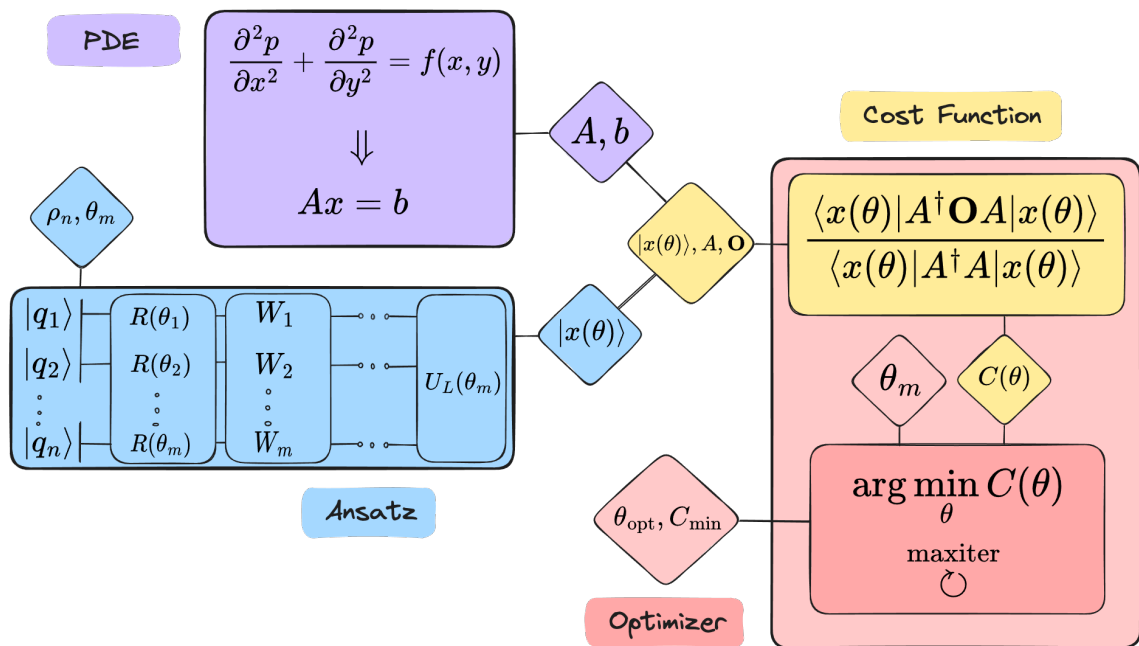


VQLS Basics



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Chapter 1

Introduction

This document covers variational quantum linear system algorithms where the goal is to take a sparse matrix A and a vector $\mathbf{b}$, and prepare a quantum state $|x\rangle$ that is proportional to $A^{-1}|b\rangle$.

I hope the reader is familiar with undergraduate linear algebra, systems of equations, and can grasp general quantum mechanics concepts. For those with a programming background and not physics, only a handful of simple concepts from quantum mechanics are required. Including dirac notation, expectation values, and general grasp of laws of superposition and entanglement. Even so, I would encourage the inexperienced reader to dive into the document. I have provided lots of diagrams, equations, and graphs to help aid the learning process.

Note if your goal is to achieve a "quantum advantage" by solving the Poisson problem using a VQLS then this project is not for you. In the end, this project amounts classically to a small to medium sized matrix multiplication which can be handled by your smart phone in a matter of milliseconds to seconds. A true advantage may be proposed for quantum systems in which a classical computer cannot handle naturally. However, I am interested in solving PDEs right now on noisy quantum hardware. So my research is more concerned with improving the scalability and runtime issues of VQLS.

Chapter 2

What is VQLS?

2.1 VQLS as a type of VQA

2.2 What are the goals of this project?

Chapter 3

The variational principle

Chapter 4

The heart of the VQLS: the
parameterized quantum circuit

Chapter 5

Building a VQLS

5.1 Constructing the operators

5.2 Choosing an ansatz

5.3 Building the cost function

5.4 Choosing an optimizer

Chapter 6

The readout problem to be continued